

Peak Detection using Transformers

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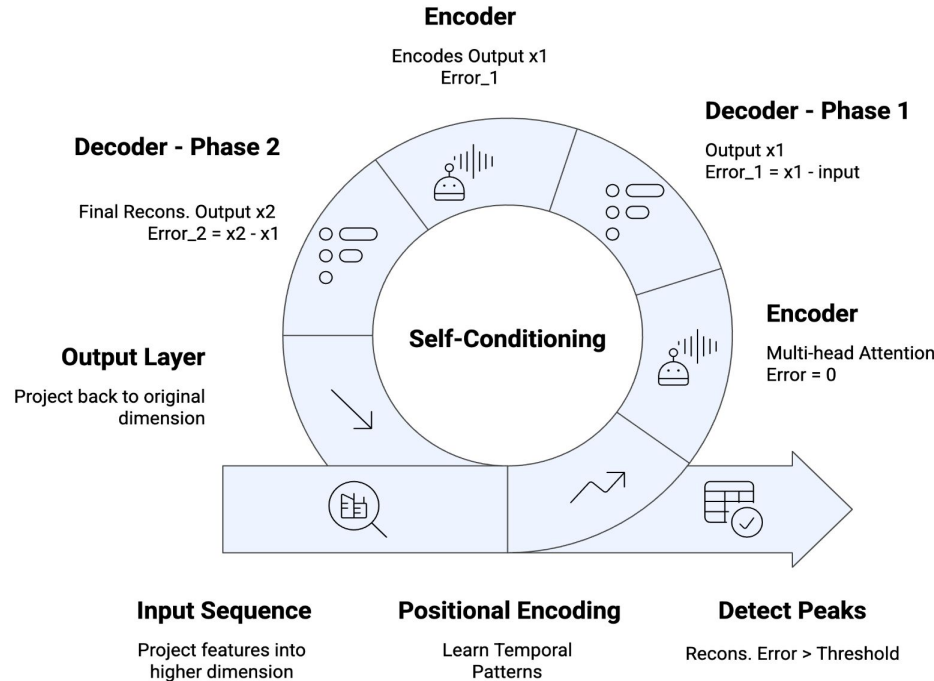


Fig : TranPD Model

- Peak = Identifying local maxima or minima
- 2017 Paper - **Attention Is All You Need!!!**
- Powers models like BERT, GPT, ViT etc.
- NLP, Computer Vision, Time-Series Forecasting

Tran-PD: Peak Detection via Reconstruction Error

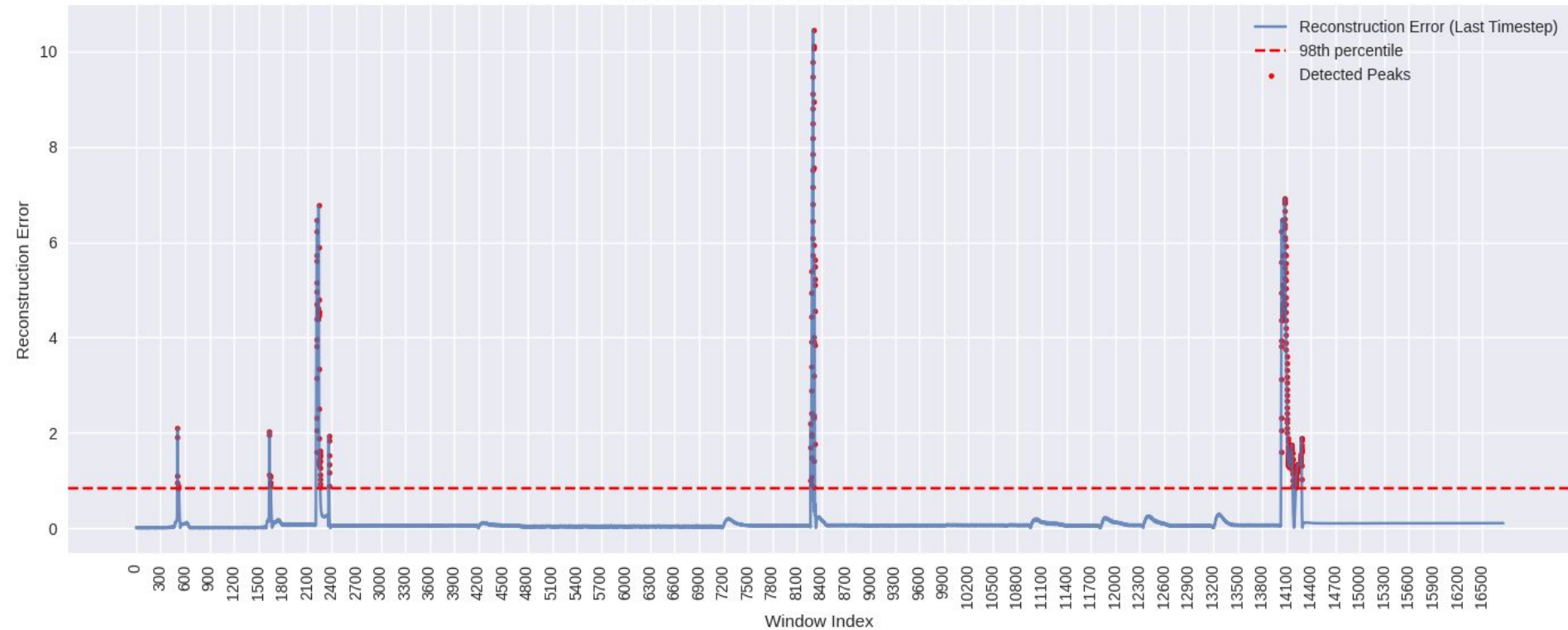


Fig : Reconstruction Error Plot - **336 Peaks Detected**

Tran-PD: Peak Detection via Reconstruction Error

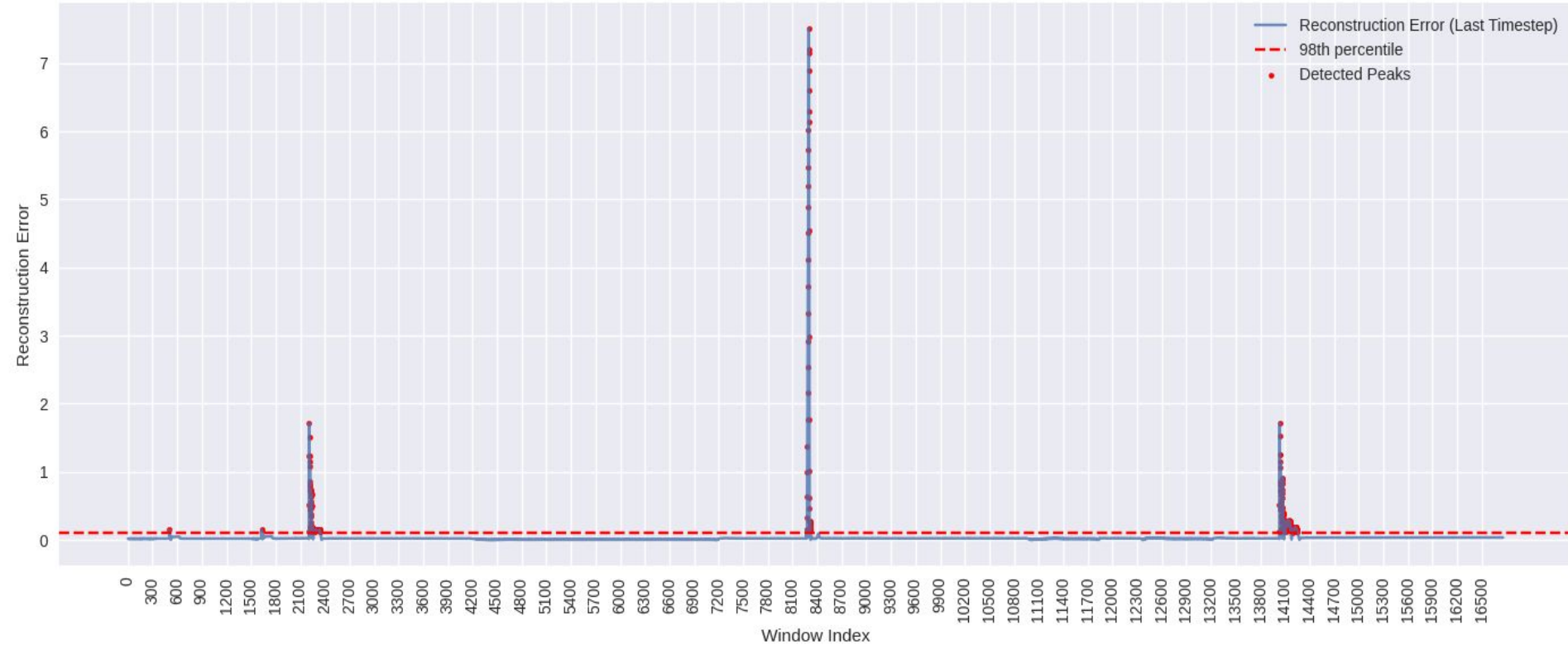


Fig : Reconstruction Error Plot - **336 Peaks Detected**

Tran-PD: Peak Detection via Reconstruction Error

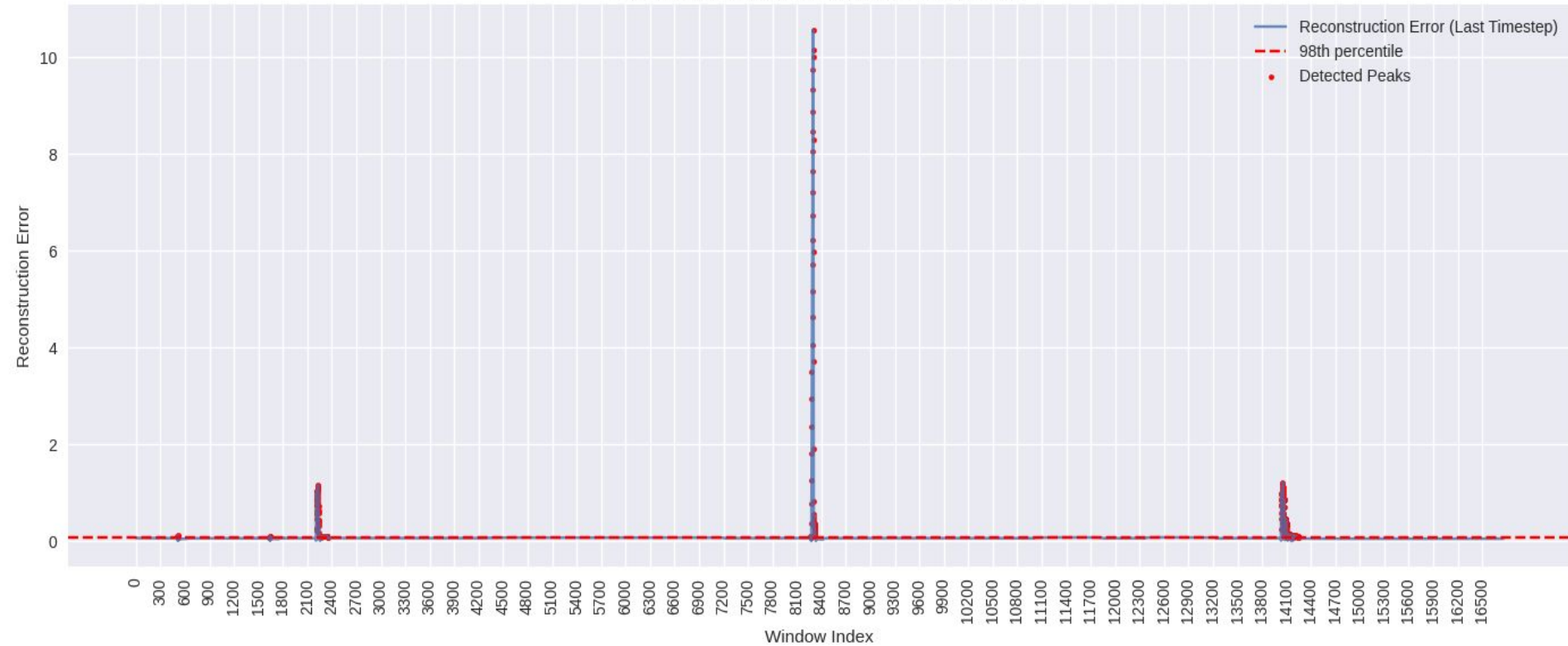


Fig : Reconstruction Error Plot - **336 Peaks Detected**

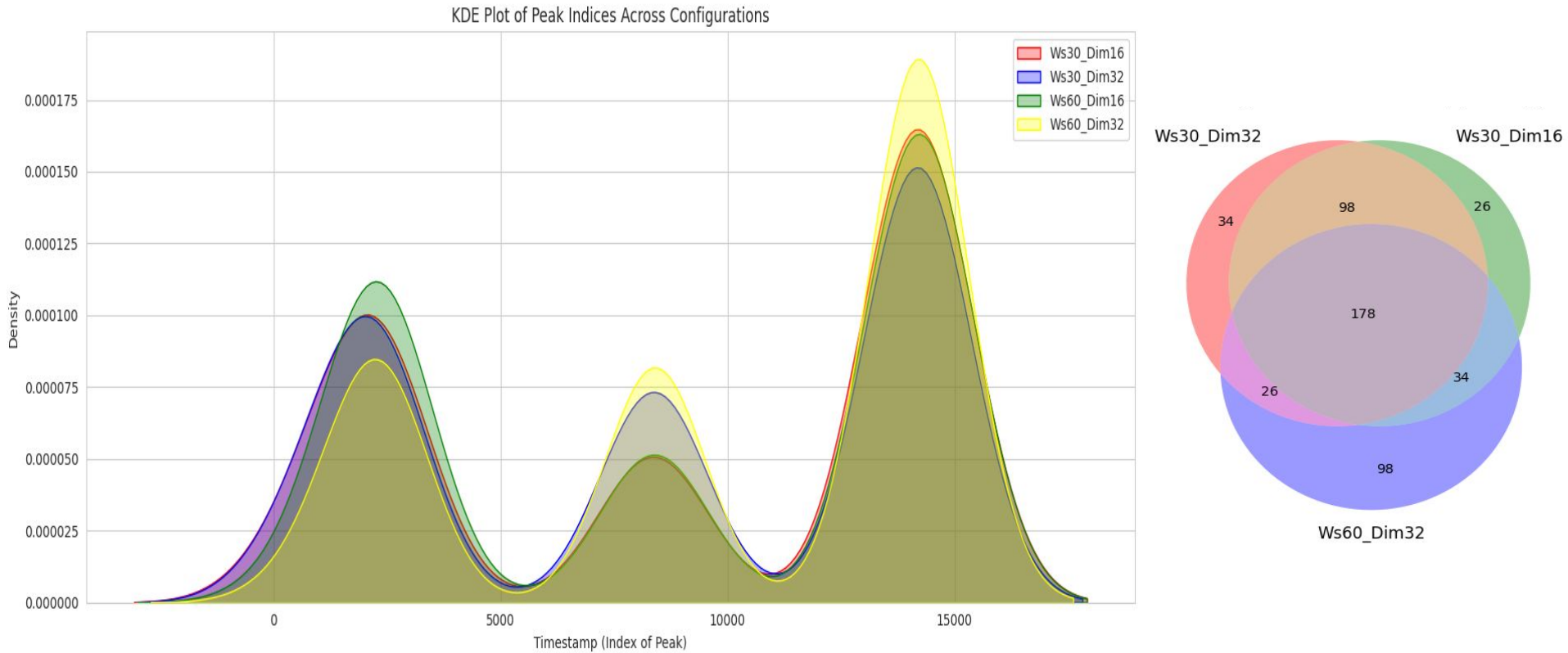


Fig : Peak Analysis Across Configurations

Measure	Peak	Non-Peak
Count	336	16447
Mean	0.74	0.11
Min	0.15	0.00
Max	8.55	0.15

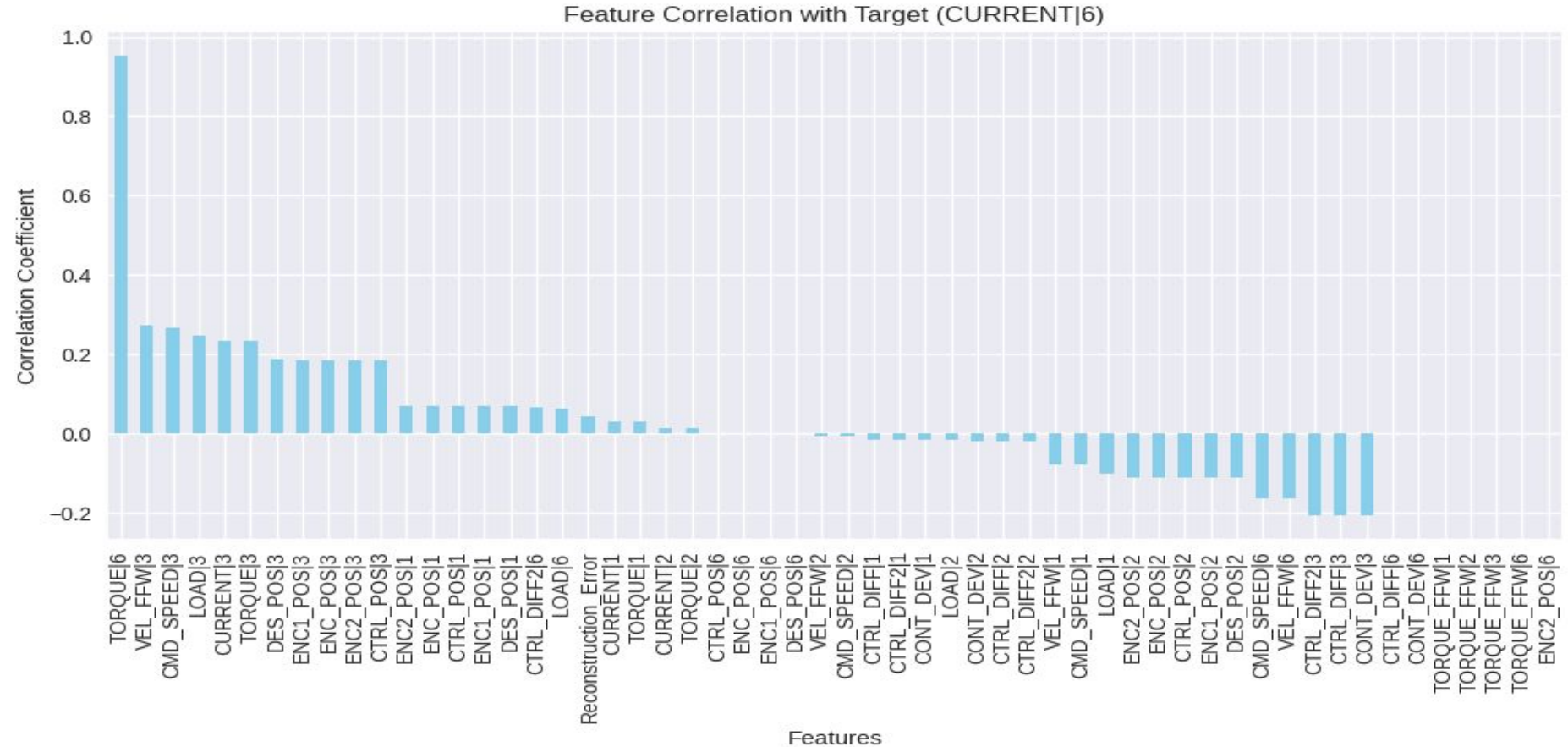
Fig : Statistical Analysis of Reconstruction Error

Measure	W = 120, D=64	W = 120, D=32	W = 120, D=16	W = 60, D=32	W = 60, D=16	W = 30, D=32	W = 30, D=16
No. of Peaks	336	336	336	336	336	336	336
Threshold	0.82	0.11	0.08	0.23	0.15	0.17	0.06
Mean RE	0.12	0.03	0.07	0.12	0.13	0.03	0.02
Min RE	0	0.00	0	0	0	0	0 - perfect zero
Max RE	10.44	7.50	10.55	8.558	8.559	9.33	9.40

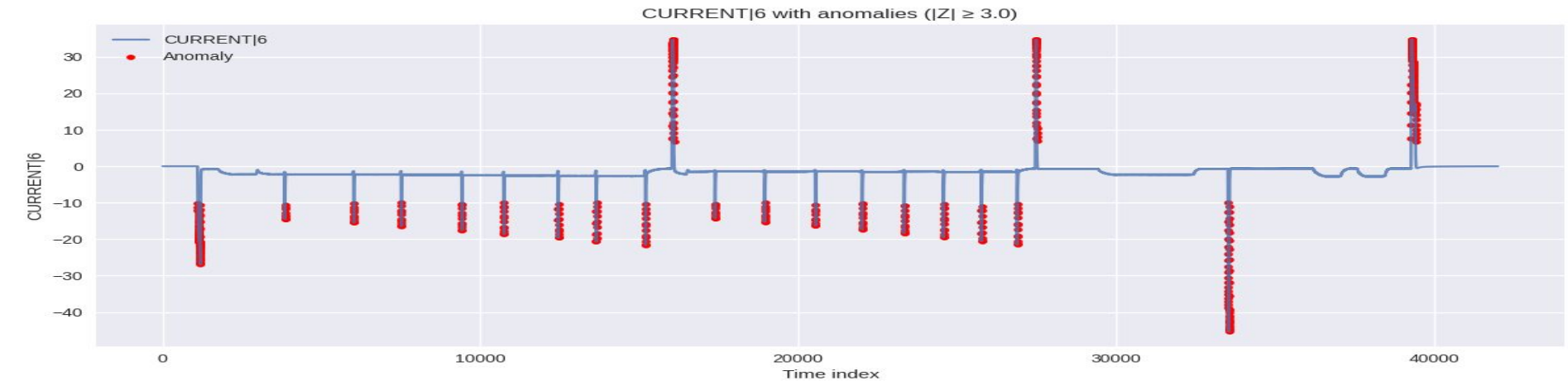
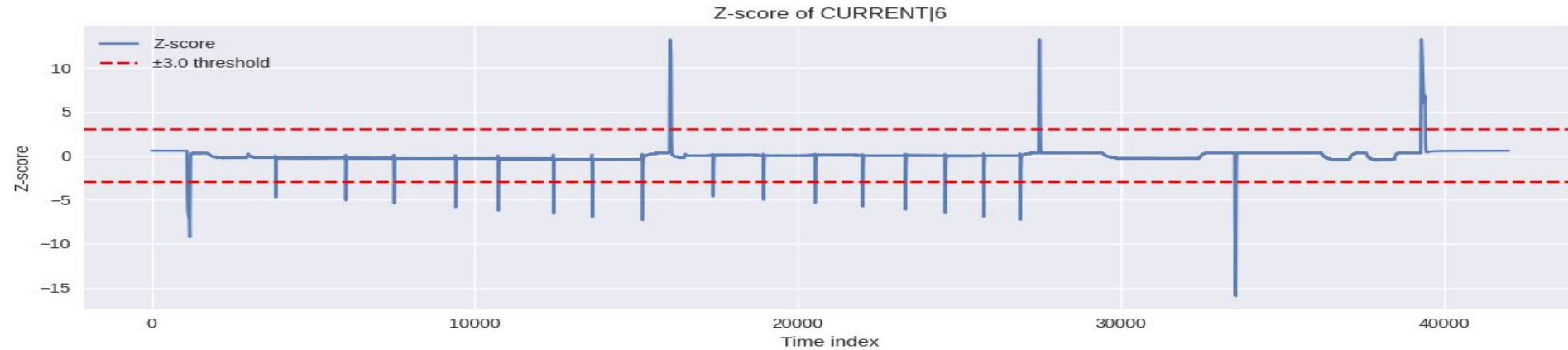
Table : Statistical Analysis of Reconstruction Error

Train Test Split	Training : Reconstruction	Random Forest with Loss	Random Forest without Loss	MAE, R ²
Chronologica	W_s = 60	Num of Features = 53	Num of Features = 52	Comparing Accuracy for Both.
	Dim = 16	Target = Current 6	Target = Current 6	
Train : Test = 25173 : 16783	Trained Model with Training Set	n_estimators = 100	n_estimators = 100	
	Reconstruction on Training Set + Reconstruction on Test Set			
	RE for complete Dataset			

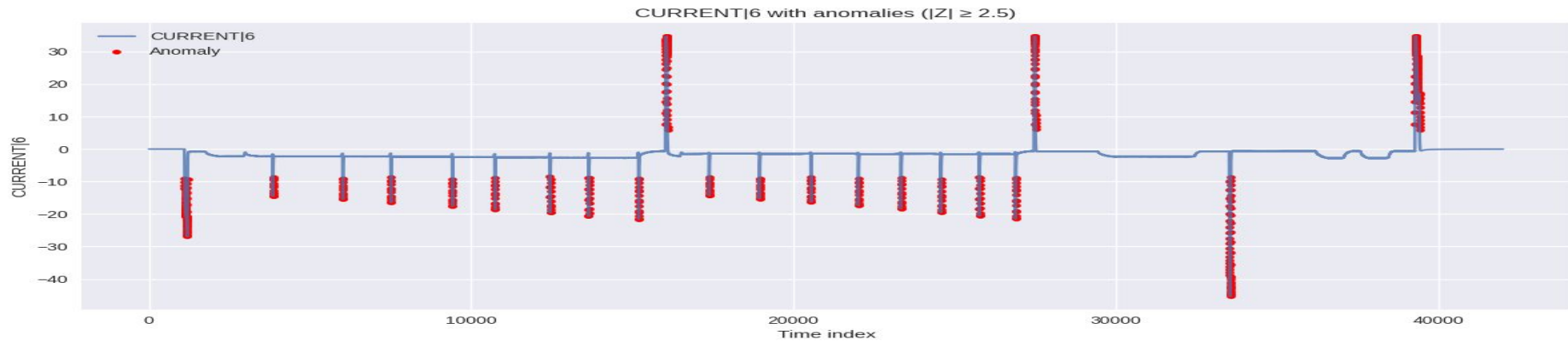
	Num of Features	MAE	RMSE	R^2
RF with Loss	53	0.1128	0.8497	0.9340
RF without Loss	52	0.1244	0.8379	0.9358



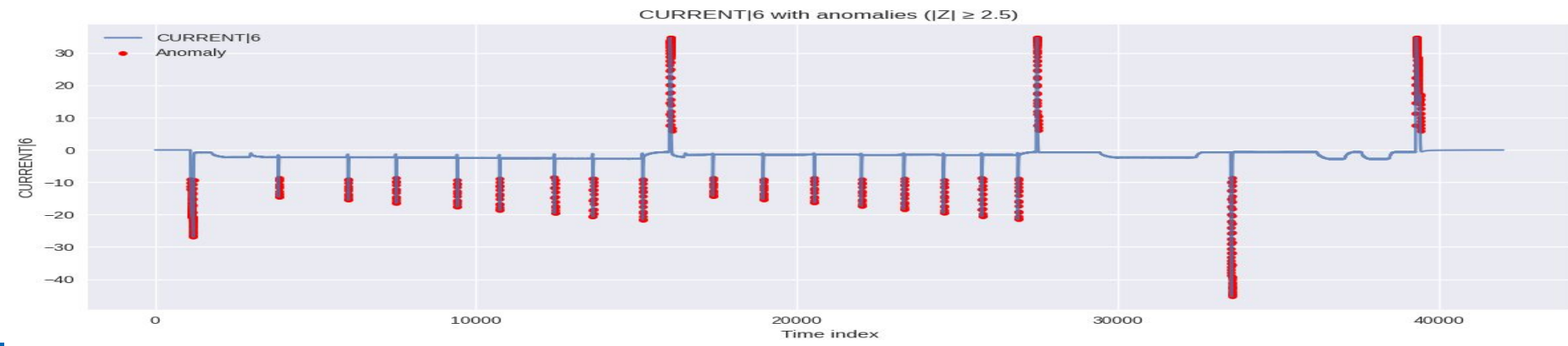
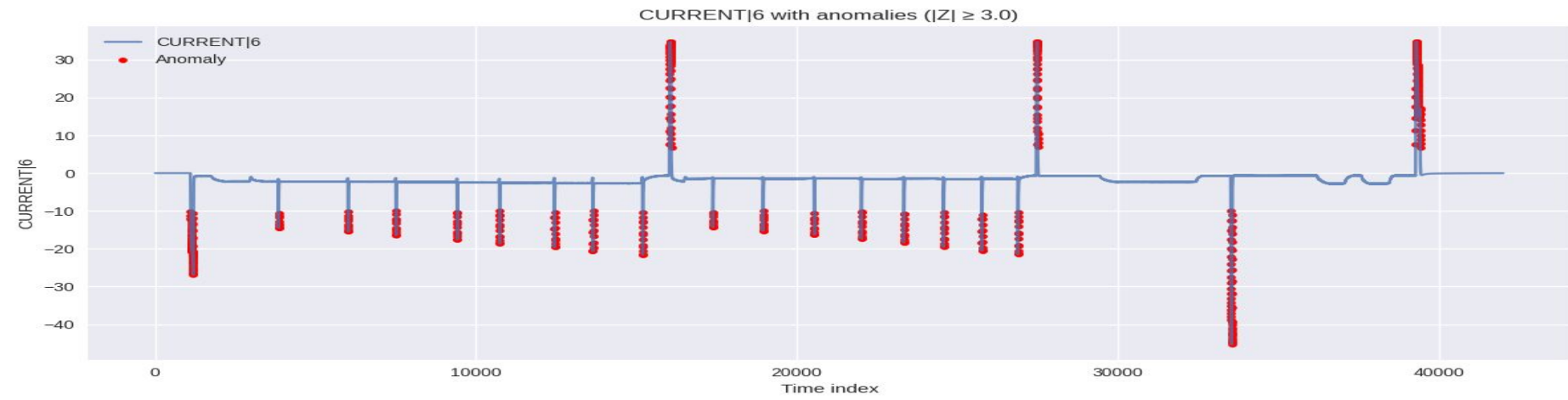
Z-Score : Threshold - 3, Num of Peaks = 548



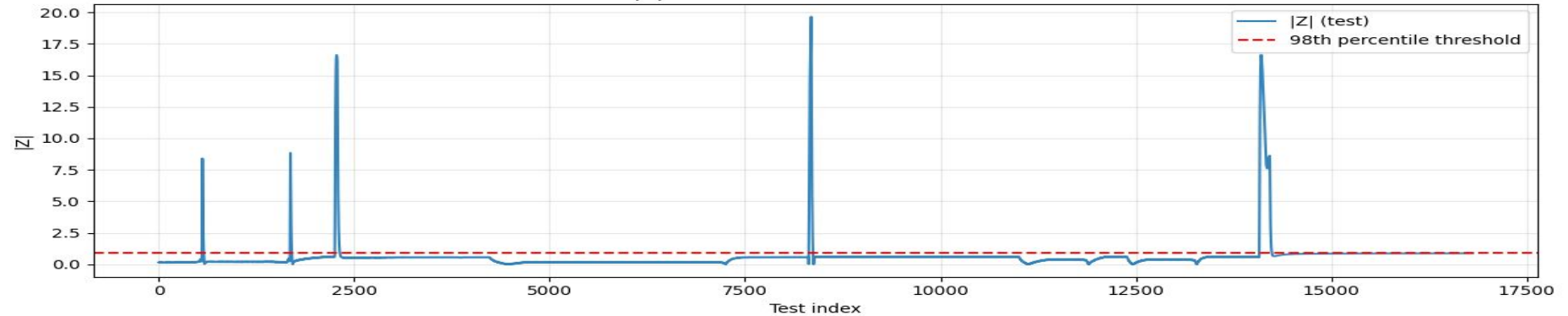
Z-Score : Threshold - 2.5, Num of Peaks = 586



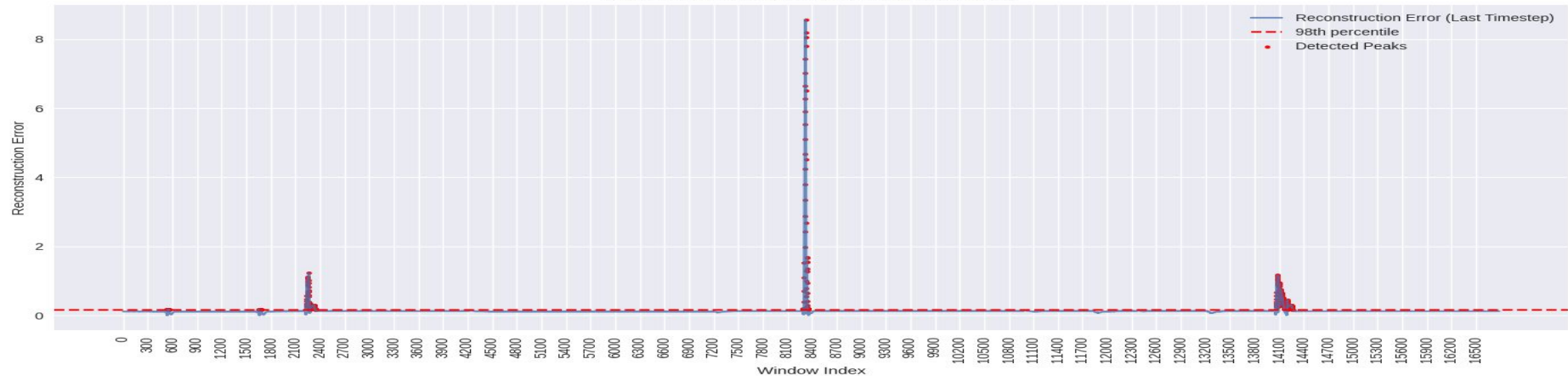
Z-Score : Threshold - 3 Vs. Threshold - 2.5



$|Z|$ Test with Percentile Threshold



Tran-PD: Peak Detection via Reconstruction Error



	Z-Score	TransformerPD
No of Peaks - Test Dataset	396	336
Mean	Z-Score : 0.6360	RE Mean - 0.13
Threshold 98%	Z-score Threshold - 0.8816	RE Threshold - 0.1565

1. Regression Analysis
2. Implement Adaptive Thresholding Mechanism
3. Multivariate Peak Detection
4. Peak Forecasting
5. regression using Loss and without Loss try to predict current and later on all cure=rent features
6. z score vs DLM - Prediction power (R^2)

1. S. Tuli, G. Casale, and N. R. Jennings, "TranAD: Deep Transformer Networks for Anomaly Detection in Multivariate Time Series Data," *Proc. VLDB Endowment (PVLDB)*, 2022. Available: <https://doi.org/10.48550/arXiv.2201.07284>
2. Z. Z. Darban, G. I. Webb, S. Pan, C. C. Aggarwal, and M. Salehi, "Deep Learning for Time Series Anomaly Detection: A Survey," *arXiv preprint arXiv:2211.05244*, 2024]. Available: <https://doi.org/10.48550/arXiv.2211.05244>
3. <https://github.com/imperial-qore/TranAD>
4. <https://www.geeksforgeeks.org/data-analysis/peak-signal-detection-in-real-time-time-series-data/>
5. <https://stackoverflow.com/questions/22583391/peak-signal-detection-in-realtime-timeseries-data>



Thank You For Your Attention!

